

Math 456 Team Essay: Forecasting GLD Using an ARMA-GJR-GARCH Model

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Introduction

This project studies the daily price behavior of GLD, the SPDR Gold Shares exchange-traded fund, using an ARMA-GJR-GARCH time-series model. GLD was selected because it is a widely traded gold-related financial asset, and gold prices often experience periods of changing volatility. This makes GLD appropriate for an ARCH/GARCH-style project, as these models are designed for time series in which the variance is not constant over time.

The goal of the project is to model and forecast GLD daily returns and volatility. Instead of modeling the raw adjusted closing price directly, the analysis converts prices into daily log returns. This is important because financial prices are often nonstationary, whereas returns are more likely to be stationary and better suited to volatility modeling. The ARMA part of the model describes the conditional mean of the return process, while the GJR-GARCH part models the conditional variance. The GJR-GARCH model is especially useful because it allows positive and negative return shocks to affect future volatility differently.

The analysis follows the same general approach as the ARMA-GARCH reference article by Ghani and Rahim, which models financial-style daily price data by converting prices into returns, testing stationarity, checking for ARCH effects, comparing candidate ARMA-GARCH specifications, and evaluating forecasting performance. In this project, the final selected model was an ARMA(0,0)-GJR-GARCH(1,1) model with a Student-t error distribution. The final model was chosen after comparing multiple candidate models using AIC and BIC, while also prioritizing the GJR-GARCH structure required for this assignment.

The R analysis used the packages quantmod, tidyverse, lubridate, tseries, FinTS, forecast, rugarch, PerformanceAnalytics, knitr, zoo, and xts. The quantmod package was used to download GLD historical data from Yahoo Finance. The tidyverse and lubridate packages were used for data cleaning, transformation, and plotting. The tseries and FinTS packages were used for stationarity and ARCH tests. The forecast package was used for ARMA mean-model selection, and rugarch was used to fit, compare, diagnose, and forecast GARCH models.

The final results showed strong evidence of volatility clustering in GLD daily returns. The ADF test supported stationarity of returns, while the Ljung-Box test on squared returns and the ARCH LM test strongly supported the presence of ARCH effects. The final

ARMA(0,0)-GJR-GARCH(1,1) Student-t model passed the main residual diagnostic checks, suggesting that it captured much of the remaining autocorrelation and volatility structure. However, the price-level forecasts were less accurate during the test period because GLD experienced a large upward movement from 2022 to 2026, which a return-based GARCH model does not fully capture since GARCH does not model the long-term direction or trend of prices.

Data Description

The dataset used in this project consists of daily GLD historical market data downloaded from Yahoo Finance. The data covers the period from January 2, 2014, through April 29, 2026. The original data contained 3,099 daily price observations. Each row represents one trading day and includes the variables date, open, high, low, close, volume, and adjusted close. The adjusted close variable was used as the main price variable because it is the standard cleaned price field used for historical return calculations.

The raw data was checked for missing values, duplicate dates, and invalid observations. There were no missing values and no duplicate dates in the downloaded dataset. The final cleaned price dataset, therefore, contained 3,099 price rows. Daily log returns were then calculated using adjusted close prices according to the formula:

$$\text{Daily log return} = 100 \times \ln(\text{Adjusted Close}_t / \text{Adjusted Close}_{(t-1)})$$

Multiplying by 100 expresses returns in percentage points. Because the first day does not have a previous price from which to calculate a return, the final return dataset contained 3,098 daily log-return observations. The mean adjusted closing price was 169.47, the minimum adjusted closing price was 100.50, and the maximum adjusted closing price was 495.90. The mean daily log return was 0.0408%, and the standard deviation of daily log returns was 0.9961%.



Figure 1: Adjusted closing price of GLD from 2014 to 2026. The plot shows that GLD was relatively stable for several years, began rising more noticeably after 2019, and increased sharply during the later part of the sample, especially around 2024–2026.

Figure 1 shows why GLD is a useful financial time-series example. The price series exhibits a clear upward trend and is non-stationary in levels, motivating the use of log returns for modeling. Because the price itself is not centered around a stable mean, the model is built on daily log returns rather than the raw price level.

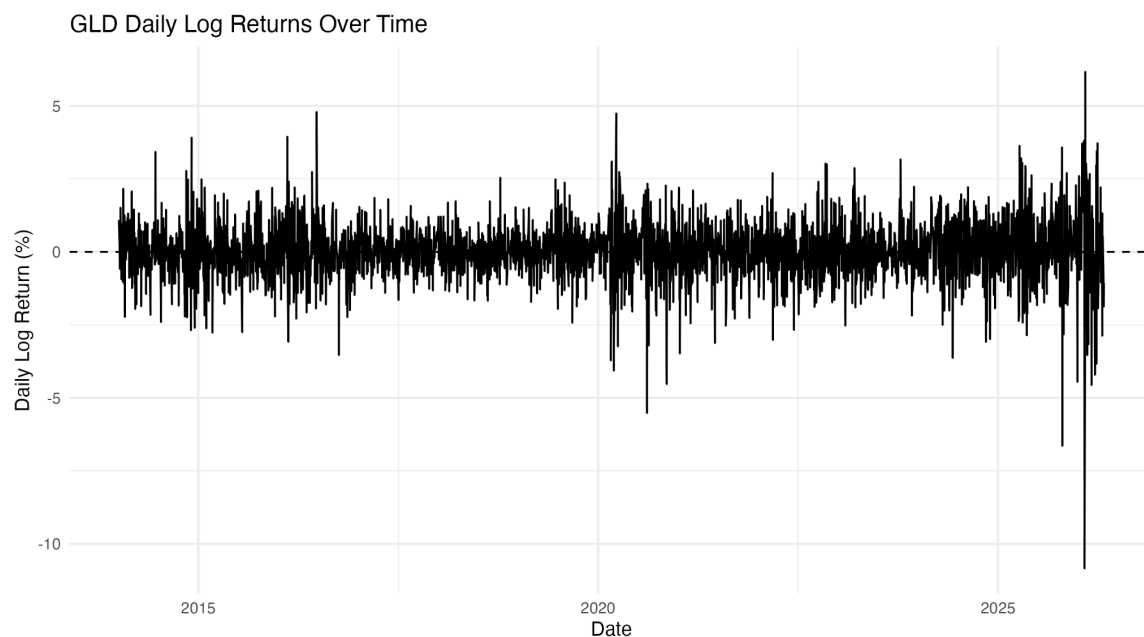


Figure 2: Daily log returns of GLD from 2014 to 2026. The returns fluctuate around zero but show clusters of high and low volatility.

Figure 2 shows the daily GLD log returns. Most returns are close to zero, but there are periods where large positive and negative returns appear close together. This pattern is called volatility clustering. Volatility clustering is one of the main reasons for using ARCH/GARCH models, because these models allow the conditional variance of returns to change over time instead of assuming one constant variance.

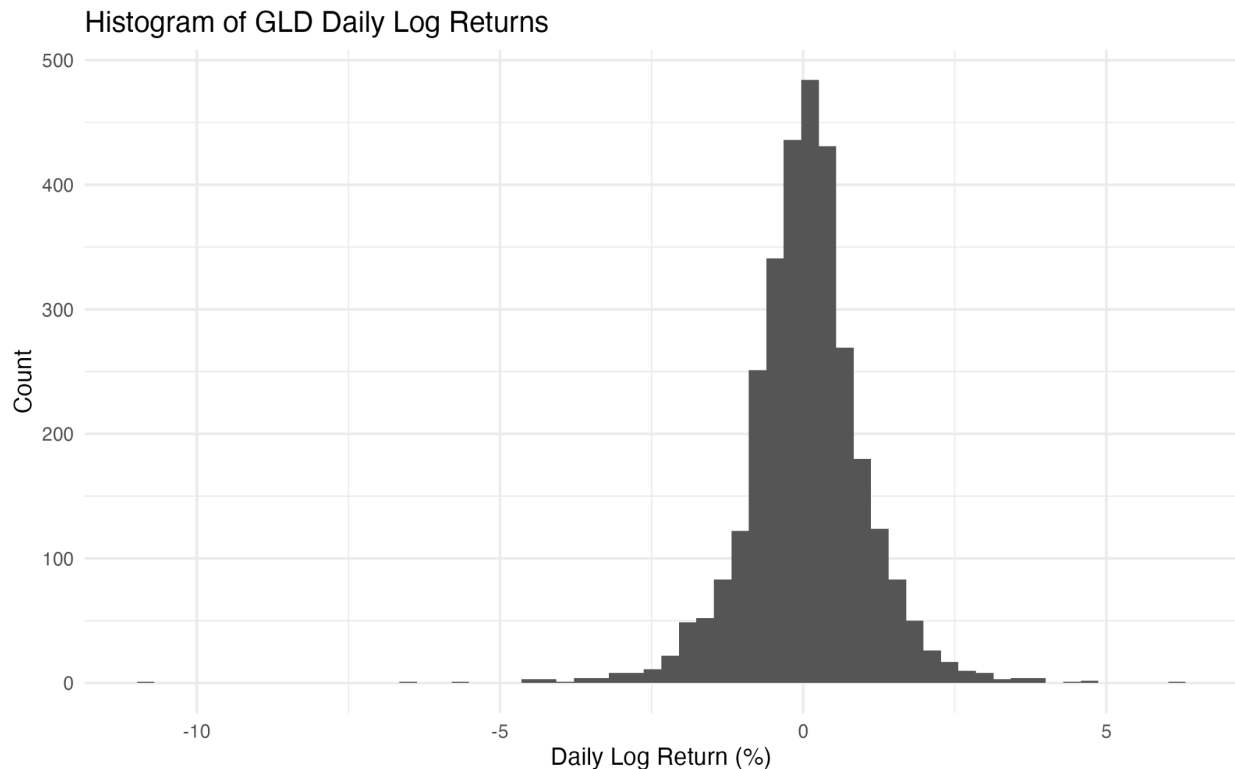


Figure 3: Histogram of GLD daily log returns. The distribution is centered near zero but has heavier tails than a normal distribution.

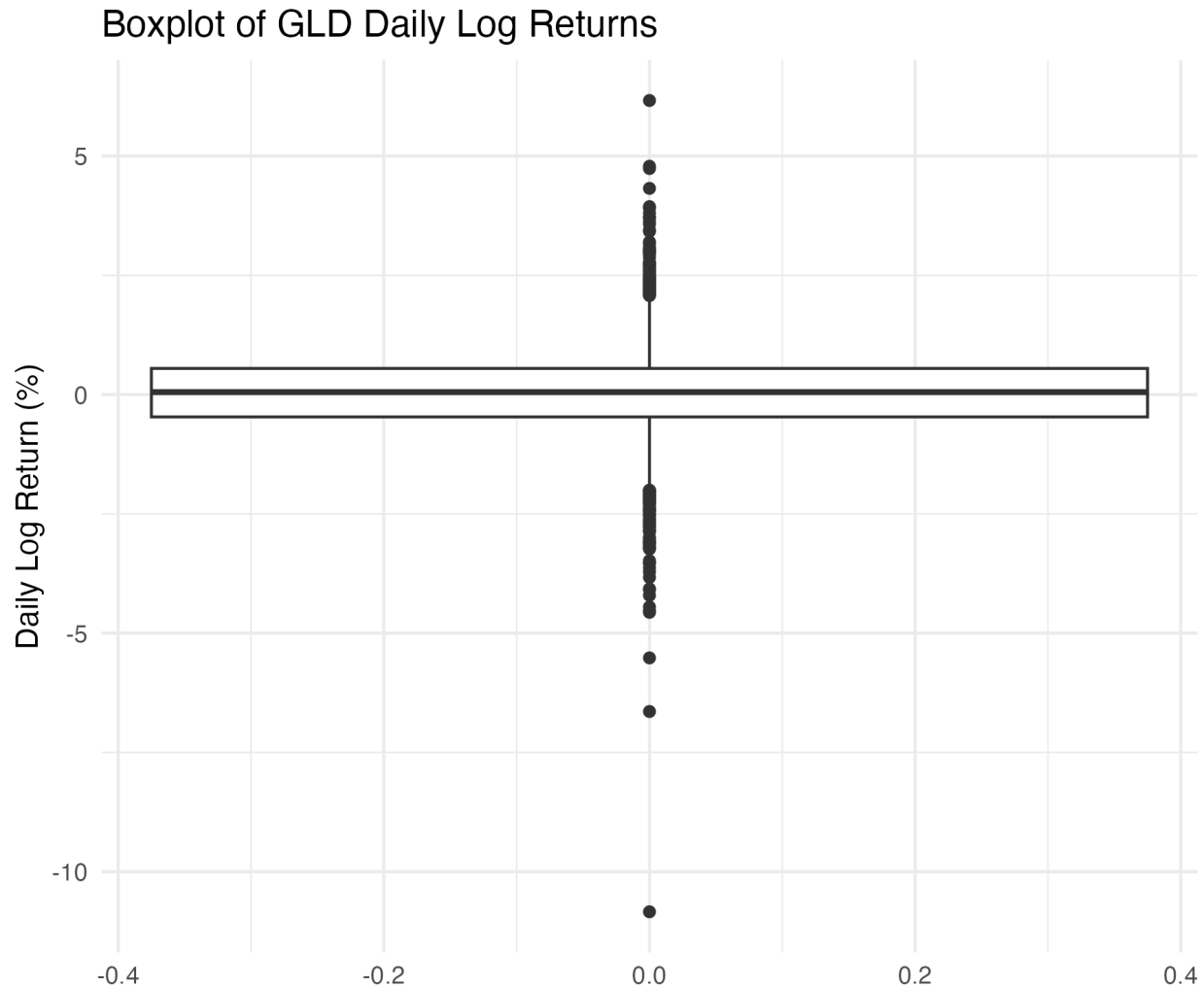


Figure 4: Boxplot of GLD daily log returns. The boxplot shows several extreme return observations on both the positive and negative sides.

The histogram and boxplot show that the return distribution is centered near zero but contains heavy tails and extreme observations. Using Tukey's IQR rule, 144 return observations were flagged as possible outliers. However, these observations were not automatically removed from the main model because extreme return days in financial markets are often real market events rather than data-entry mistakes. Since GARCH models are specifically designed to model changes in volatility, removing all large return days would risk removing the exact behavior the model is supposed to capture. Instead, missing and invalid data were removed, outliers were flagged, and a winsorized sensitivity version of the return dataset was created if stricter outlier handling is required later.

Analysis

After cleaning and return construction, the GLD return dataset was split into training and test sets. Because this is time-series data, the split was chronological rather than random. This prevents future information from leaking into the training process. The first 70% of observations were used as the training set, and the remaining 30% were used as the test set. The training set contained 2,168 observations from January 3, 2014 through August 12, 2022. The test set contained 930 observations from August 15, 2022 through April 29, 2026.

Table 1. Train-test split sizes

Dataset	Rows	Start Date	End Date
Training set	2,168	2014-01-03	2022-08-12
Test set	930	2022-08-15	2026-04-29

The first major modeling requirement was to check whether the return series was appropriate for time-series modeling. The Augmented Dickey-Fuller test was used to test stationarity. The ADF statistic was -12.396 with a p-value of < 0.01 , which supports rejecting the null hypothesis of a unit root. This means the GLD return series can be treated as stationary for the purpose of ARMA-GARCH modeling.

Next, the Ljung-Box test was used to check whether there was autocorrelation in the return series. The Ljung-Box test on returns had a statistic of 32.588 and a p-value of 0.0374, suggesting mild autocorrelation in the mean process. However, the ARMA model selection step later selected ARMA(0,0), meaning that adding AR or MA terms did not improve the model enough under the BIC criterion. This suggests that the mean return process is close to white noise.

The most important tests for this project were the squared-return Ljung-Box test and ARCH LM test. The Ljung-Box test on squared returns had a statistic of 325.354 and a p-value reported as 0, while the ARCH LM test had a statistic of 127.707 and a p-value of 1.78×10^{-21} . These results strongly reject the null hypothesis of no ARCH effects. In other words, the variance of GLD returns is not constant over time (evidence of conditional heteroskedasticity), and past volatility helps explain future volatility. This provides strong evidence that a GARCH-type model is appropriate.

Table 2. Preliminary diagnostic tests

Test	Statistic	p-value	Interpretation
ADF stationarity test	-12.396	0.010	Returns are stationary
Ljung-Box returns	32.588	0.037	Mild mean autocorrelation
Ljung-Box squared returns	325.354	< 0.001	Strong volatility dependence
ARCH LM test	127.707	1.78e-21	Strong ARCH effects

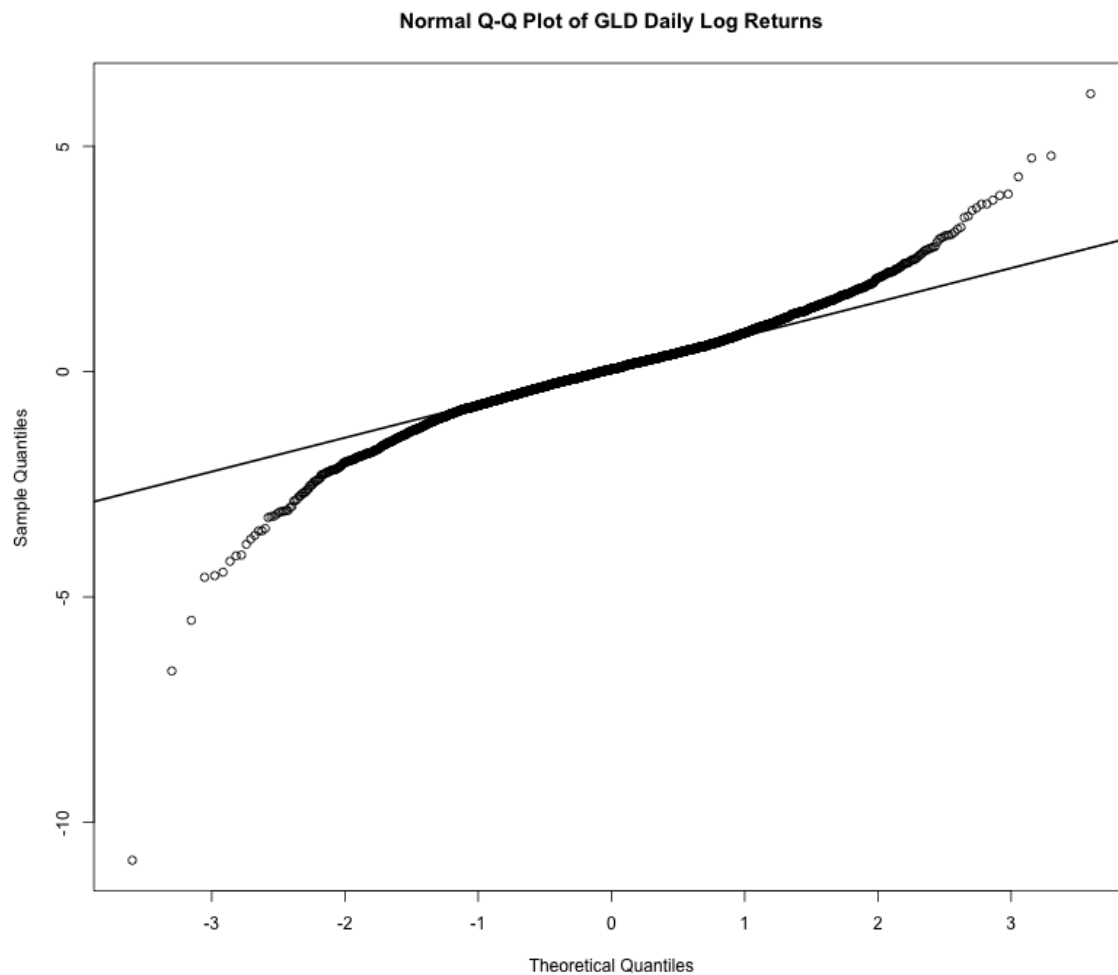


Figure 5: Normal Q-Q diagonal plot of GLD daily log returns. The plot shows that the central part of the return distribution follows the diagonal more closely, while the tails deviate from the line.

The required diagonal plot is shown in Figure 5. A normal Q-Q plot compares the observed return distribution with what would be expected under a normal distribution. The points

deviate from the diagonal line in the tails, showing that GLD returns have heavier tails than a normal distribution. This supports using a Student-t error distribution in the GARCH model rather than assuming normally distributed errors.

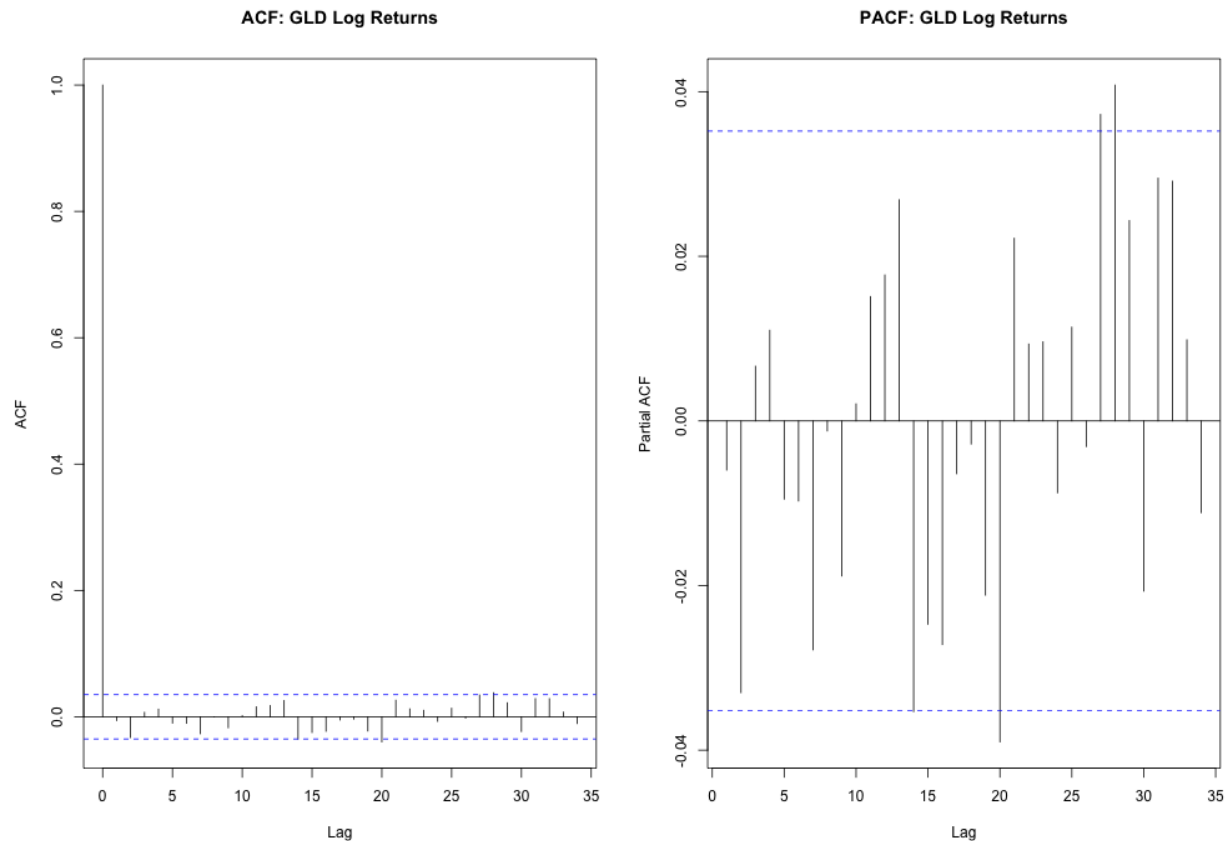


Figure 6: ACF and PACF plots of GLD daily log returns. The plots show limited autocorrelation in the return series, supporting a simple mean model.

Figure 6 shows the ACF and PACF of daily log returns. There is not a strong pattern of autocorrelation in returns, which supports the ARMA(0,0) mean model selected by BIC. This means that the best mean model is essentially a constant mean return rather than a model with meaningful autoregressive or moving-average structure.

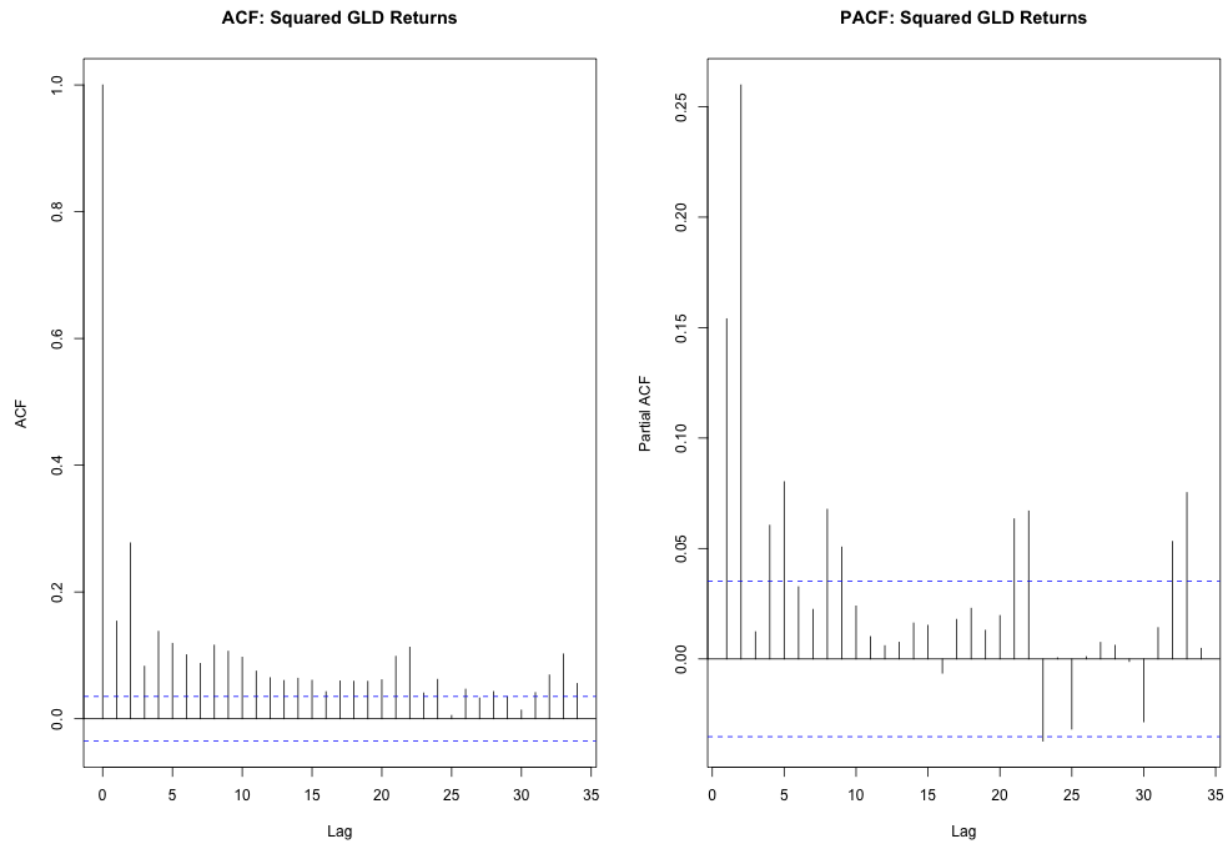


Figure 7: ACF and PACF plots of squared GLD daily log returns. The plots show dependence in squared returns, which supports the use of a GARCH-type volatility model.

Figure 7 shows the ACF and PACF of squared returns. Squared returns are used as a proxy for volatility, so autocorrelation in squared returns indicates volatility clustering. The visible dependence in squared returns agrees with the ARCH LM test and supports the need for a model with time-varying conditional variance.

Candidate GARCH models were then compared using AIC and BIC. The models considered included standard GARCH and GJR-GARCH variance specifications, ARMA(0,0), ARMA(1,0), ARMA(0,1), and ARMA(1,1) mean structures, and normal, Student-t, and GED error distributions. The lowest BIC overall was obtained by a standard GARCH(1,1) model with ARMA(0,0) and Student-t errors. However, the best-performing GJR-GARCH specification had a comparable BIC and was selected as the final model due to the requirement to use a GJR-GARCH framework in this assignment. The final selected model was ARMA(0,0)-GJR-GARCH(1,1) with Student-t errors.

Table 3. Top candidate GARCH models by BIC

Rank	Variance Model	ARMA Order	Distribution	AIC	BIC
1	sGARCH	ARMA(0,0)	Student-t	2.4485	2.4616
2	gjrGARCH	ARMA(0,0)	Student-t	2.4459	2.4617
3	sGARCH	ARMA(0,1)	Student-t	2.4491	2.4648
4	sGARCH	ARMA(1,0)	Student-t	2.4491	2.4648
5	gjrGARCH	ARMA(0,1)	Student-t	2.4466	2.4649

The final model can be summarized as:

Mean model: ARMA(0,0)

Variance model: GJR-GARCH(1,1)

Error distribution: Student-t

The ARMA(0,0) part means that the daily mean return is modeled as a constant. The GJR-GARCH(1,1) part means that today's volatility depends on a long-run volatility term, yesterday's squared shock, yesterday's conditional variance, and an asymmetric shock term. The Student-t distribution was used because GLD returns showed heavy tails in the histogram, boxplot, and Q-Q plot.

Model Evaluation and Prediction

The final ARMA(0,0)-GJR-GARCH(1,1) model with Student-t errors was fitted on the training set. The model had a log-likelihood of -2645.4, AIC of 2.4459, and BIC of 2.4617. The estimated mean return parameter was 0.0269, but it was not statistically significant at the 5% level using robust standard errors. This is reasonable because daily financial returns are difficult to predict in the mean.

The variance equation parameters were more important. The omega parameter was 0.0127 and statistically significant, showing a positive baseline level of volatility. The alpha1 parameter was 0.0633 and statistically significant, meaning that recent return shocks affect current volatility. The beta1 parameter was 0.9429 and highly significant, meaning that volatility is highly persistent over time. The gamma 1 parameter was -0.0389 and statistically significant, indicating an asymmetric volatility effect. This means that negative and positive return shocks have different impacts on future volatility. The shape parameter of the Student-t distribution was 5.019 and highly significant, which supports the use of a heavy-tailed distribution.

Table 4. Final model summary

Component	Value
Final variance model	GJR-GARCH(1,1)
Final mean model	ARMA(0,0)
Error distribution	Student-t
AIC	2.4459
BIC	2.4617
Log-likelihood	-2645.4

The next step was to evaluate whether the model adequately removed the autocorrelation and ARCH effects that were present before modeling. Three residual diagnostic tests were used: the Ljung-Box test on standardized residuals, the Ljung-Box test on squared standardized residuals, and the ARCH LM test on standardized residuals. The p-values were 0.302, 0.660, and 0.722, respectively. Since all three p-values are above 0.05, there is no strong remaining evidence of serial correlation or ARCH effects in the standardized residuals. This means the final model provides a satisfactory fit to the training return data.

Table 5. Final model residual diagnostics

Test	Statistic	p-value	Interpretation
Ljung-Box standardized residuals	22.726	0.302	No strong remaining serial correlation
Ljung-Box squared standardized residuals	16.893	0.660	No strong remaining volatility clustering
ARCH LM standardized residuals	8.770	0.722	No strong remaining ARCH effects

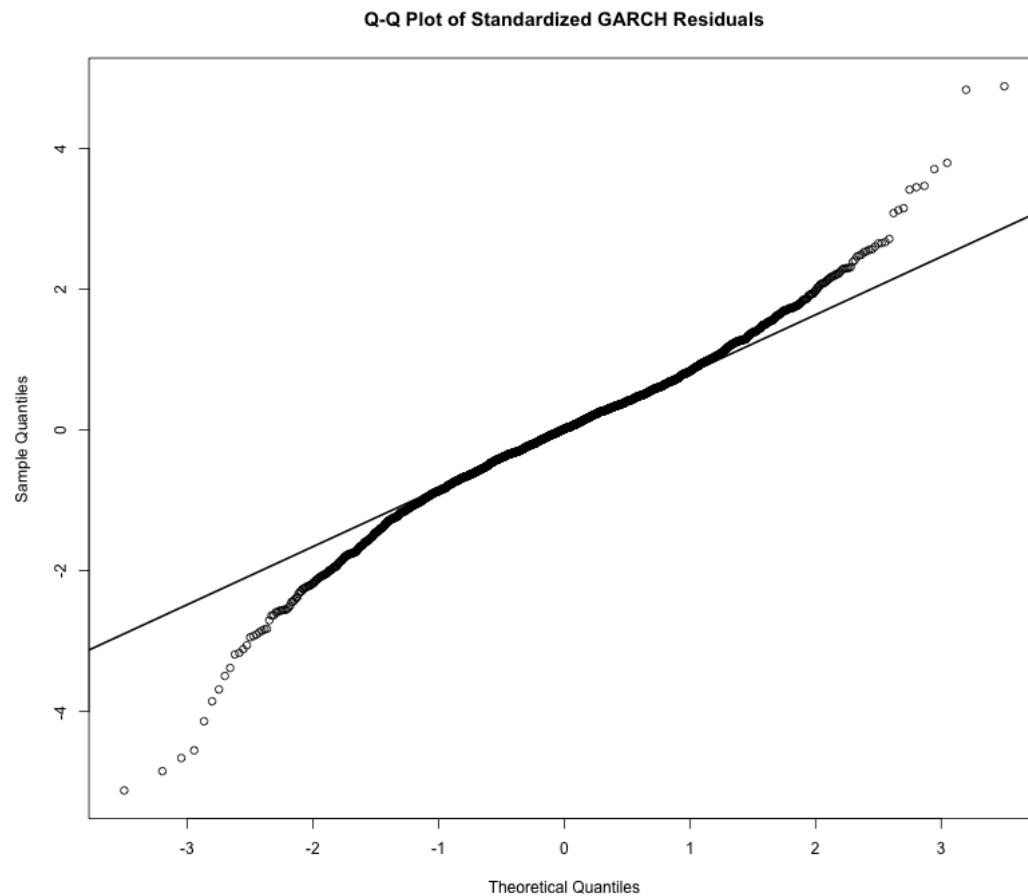


Figure 8: Q-Q plot of standardized GARCH residuals. Compared with the raw return Q-Q plot, the standardized residuals are more consistent with the fitted Student-t structure, although some tail deviations remain.

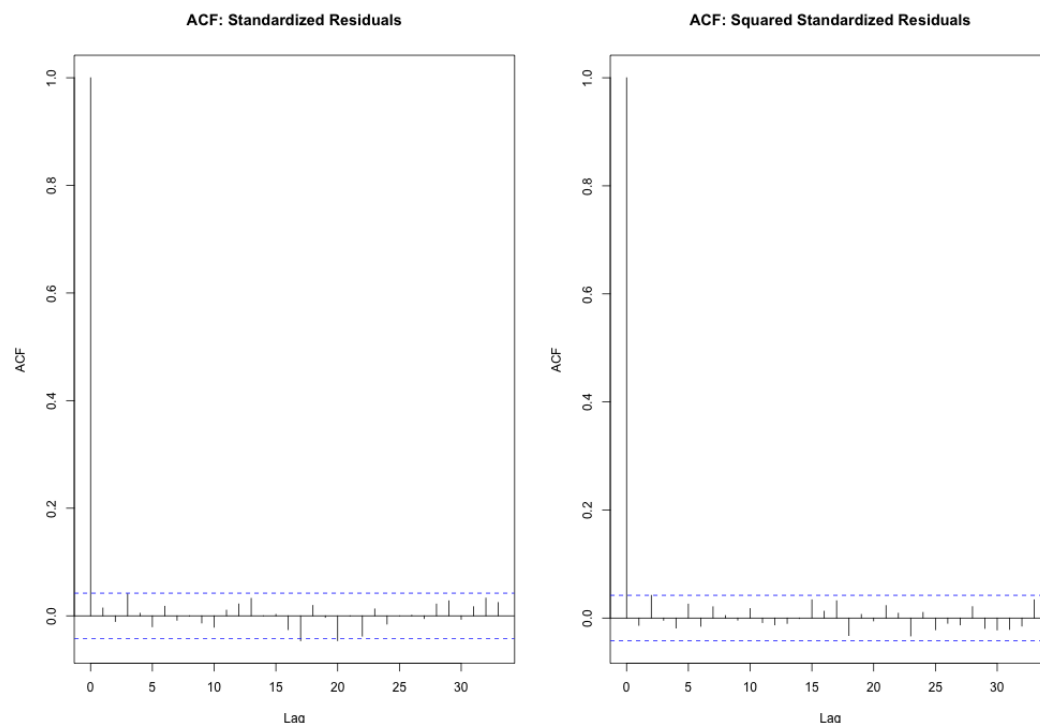


Figure 9: ACF plots of standardized residuals and squared standardized residuals. The plots show much weaker remaining autocorrelation after fitting the model.

Figures 8 and 9 support the residual diagnostic tests. The standardized residual Q-Q plot still shows some tail behavior, which is normal in financial returns, but the residual autocorrelation plots are much cleaner than the original squared-return plots. This confirms that the GJR-GARCH model captured much of the volatility clustering in the data.

The model was then evaluated on the held-out test set from August 15, 2022 through April 29, 2026. The return forecast RMSE was 1.211, and the return forecast MAE was 0.850. The direction accuracy was 54.6%, meaning the model predicted the correct direction of the daily return slightly more than half the time. The price forecast RMSE was 95.159, and the price forecast MAE was 62.486.

Table 6. Test-set forecasting performance

Metric	Value
Return RMSE	1.211
Return MAE	0.850
Price RMSE	95.159

Price MAE	62.486
Direction accuracy	54.6%
Volatility proxy RMSE	0.836

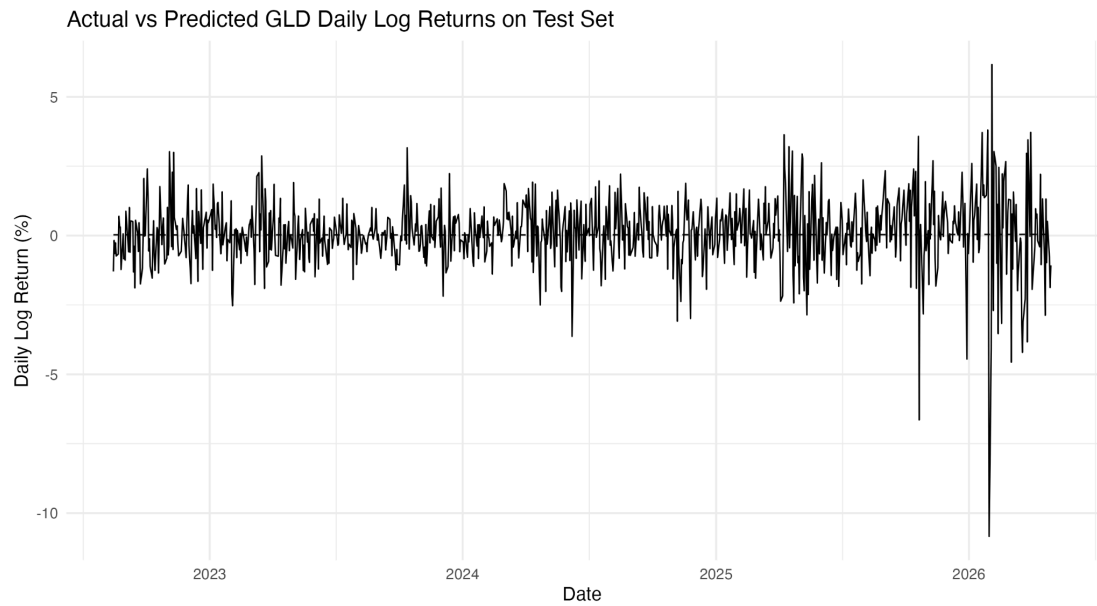


Figure 10: Actual versus predicted GLD daily log returns on the test set. The predicted returns stay close to the average daily return, while actual returns fluctuate much more.

Figure 10 shows that the model does not predict individual daily returns very strongly. This is expected because daily financial returns are noisy and difficult to forecast in direction. The ARMA(0,0) mean model predicts a nearly constant expected return, so the model is more useful for volatility forecasting than for predicting exact daily return values.

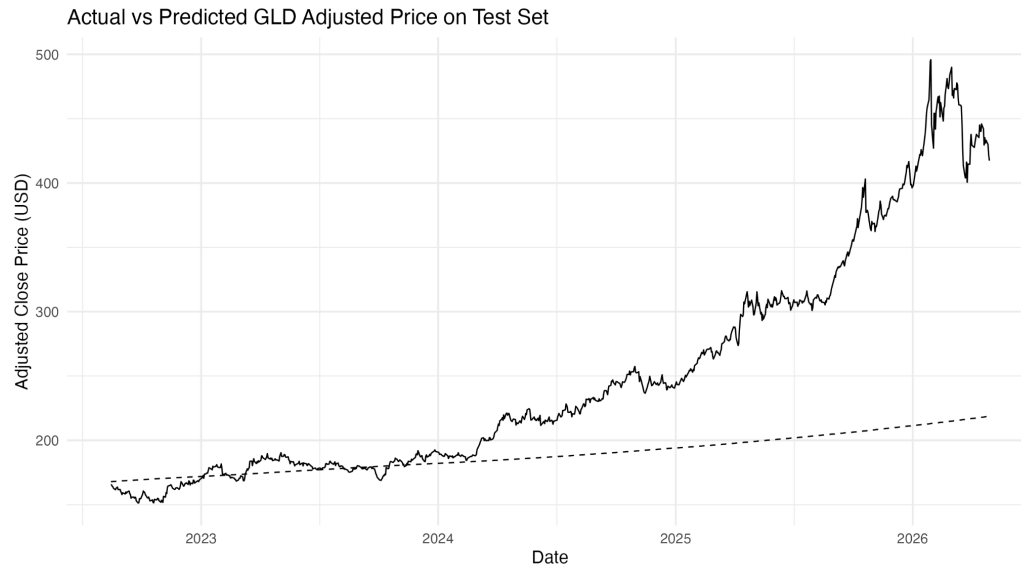


Figure 11: Actual versus predicted adjusted GLD price on the test set. The predicted price path is smoother and does not fully capture the sharp increase in GLD during the test period.

Figure 11 shows the main weakness of the model. The predicted price path is too smooth and does not follow the large increase in GLD prices from 2024 to 2026. This is because the GARCH model was built on returns and volatility, not on a structural model of gold price levels. The model can capture risk and volatility better than it can capture a major trend or regime shift in the underlying price.

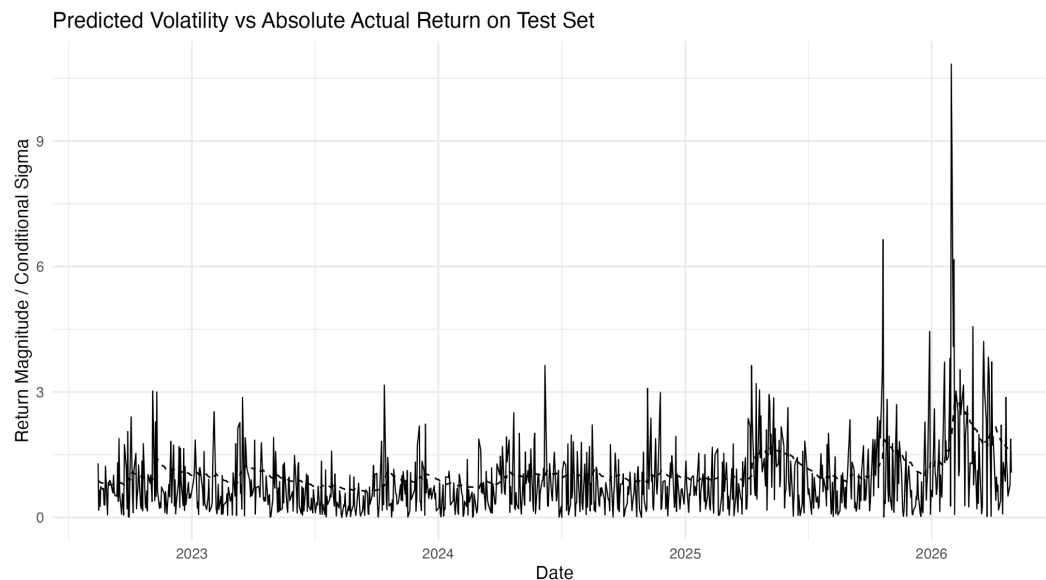


Figure 12: Predicted volatility compared with absolute actual returns. The predicted sigma series is smoother than absolute returns but reflects the model's estimate of changing conditional volatility.

Figure 12 is one of the most important prediction plots because it matches the purpose of a GARCH model. The predicted conditional standard deviation changes over time and is meant to represent expected volatility. Absolute returns are noisy day-by-day, so the model does not need to match every spike exactly. Instead, the goal is to capture periods where risk is higher or lower. The model's ability to remove remaining ARCH effects from the standardized residuals suggests that it successfully captured much of the volatility structure.

Finally, a 10-trading-day future forecast was produced. The model predicted a daily return of approximately 0.0269% for each of the next 10 trading days. The predicted conditional volatility increased slightly from 0.866% to 0.879%. The predicted adjusted price rose gradually from 417.52 on April 30, 2026 to 418.53 on May 13, 2026.

Table 7. Ten-day future forecast

Forecast Day	Approx. Trading Date	Predicted Return %	Predicted Sigma %	Predicted Adjusted Price
1	2026-04-30	0.0269	0.8657	417.52
2	2026-05-01	0.0269	0.8672	417.63
3	2026-05-04	0.0269	0.8688	417.75
4	2026-05-05	0.0269	0.8703	417.86
5	2026-05-06	0.0269	0.8718	417.97
6	2026-05-07	0.0269	0.8733	418.08
7	2026-05-08	0.0269	0.8747	418.20
8	2026-05-11	0.0269	0.8762	418.31
9	2026-05-12	0.0269	0.8776	418.42
10	2026-05-13	0.0269	0.8790	418.53

Overall, the model should be interpreted as a volatility model rather than a perfect price-prediction model. It performs well at identifying and modeling volatility clustering, but it is limited in its ability to forecast large directional price changes.

Conclusion

In this project, daily SPDR Gold Shares data from Yahoo Finance was used to build an ARMA-GJR-GARCH forecasting model. The dataset covered January 2, 2014 to April 29, 2026 and contained 3,099 price observations. After cleaning, there were no missing values or duplicates. Daily log returns were computed from adjusted close prices, producing 3,098 observations. The data was split chronologically into 2,168 training observations and 930 test observations.

Preliminary analysis showed that returns were stationary but exhibited strong volatility clustering. The ADF test supported stationarity, while the Ljung-Box test on squared returns and the ARCH LM test confirmed strong ARCH effects. Plots and summary statistics indicated heavy tails and extreme observations, supporting the use of a GARCH model with a heavy-tailed distribution.

After model comparison, the final model selected was ARMA(0,0)-GJR-GARCH(1,1) with Student-t errors. The ARMA(0,0) structure indicated little predictable mean behavior, while the GJR-GARCH model showed highly persistent and asymmetric volatility. The Student-t distribution captured heavy tails in the return distribution.

Model diagnostics showed no significant remaining autocorrelation or ARCH effects in standardized residuals (all p-values > 0.05), indicating a good fit for volatility modeling. The model also produced interpretable volatility forecasts.

However, the model performed less well in forecasting price levels, as the test period included a strong upward trend that the model could not capture. This is expected, as GARCH models focus on volatility rather than long-term price direction. Directional accuracy was 54.6%, showing limited ability to predict return direction.

The model's limitations likely arise from excluding macroeconomic drivers such as inflation, interest rates, currency movements, and geopolitical factors. Future work could incorporate such variables, compare with other gold assets, test alternative GARCH specifications, or assess robustness using winsorized returns.

Overall, the ARMA(0,0)-GJR-GARCH(1,1) Student-t model was effective for modeling volatility in GLD returns, but limited in capturing long-term price movements.

References

Ghani, I. M. Md., & Rahim, H. A. (2019). Modeling and Forecasting of Volatility using ARMA-GARCH: Case Study on Malaysia Natural Rubber Prices. IOP Conference Series: Materials Science and Engineering, 548, 012023.

Hasan, M. (2021). Microsoft Stock Price Analysis using GARCH model in R. RPubS. https://rpubs.com/Mahmud_Hasan/778532

Yahoo Finance. GLD Historical Data. <https://finance.yahoo.com/quote/GLD/history/>

Ghalanos, A. rugarch: Univariate GARCH Models. R package documentation.

SPDR Gold Shares. GLD fund information. State Street Global Advisors.

Key R code

```
None

# =====
# 0. Packages
# =====
options(repos = c(CRAN = "https://cloud.r-project.org"))
needed_packages <- c(
  "quantmod", "tidyverse", "lubridate", "tseries", "FinTS",
  "forecast", "rugarch", "PerformanceAnalytics", "knitr", "zoo", "xts"
)

installed <- rownames(installed.packages())
for (p in needed_packages) {
  if (!(p %in% installed)) install.packages(p)
}

library(quantmod)
library(tidyverse)
library(lubridate)
library(tseries)
library(FinTS)
library(forecast)
library(rugarch)
library(PerformanceAnalytics)
library(knitr)
library(zoo)
library(xts)
```

```

set.seed(456)

# =====
# 1. Download GLD daily data from Yahoo Finance
# =====
symbol <- "GLD"
start_date <- as.Date("2014-01-01")
end_date <- Sys.Date()

out_dir <- "gld_project6_outputs"
dir.create(out_dir, showWarnings = FALSE)
dir.create(file.path(out_dir, "plots"), showWarnings = FALSE)

gld_xts <- getSymbols(symbol, src = "yahoo", from = start_date, to = end_date,
auto.assign = FALSE)

gld_prices <- data.frame(date = index(gld_xts), coredata(gld_xts))
colnames(gld_prices) <- c("date", "open", "high", "low", "close", "volume",
"adjusted")

# Basic raw data checks
raw_missing_counts <- sapply(gld_prices, function(x) sum(is.na(x)))
raw_duplicate_dates <- sum(duplicated(gld_prices$date))

# Clean prices: remove missing values, duplicate dates, and zero/negative volume
days

gld_prices_clean <- gld_prices %>%
  arrange(date) %>%
  distinct(date, .keep_all = TRUE) %>%
  filter(if_all(c(open, high, low, close, volume, adjusted), ~ !is.na(.x))) %>%
  filter(volume > 0)

# Create daily log returns using adjusted close.
# Multiplying by 100 puts returns in percentage points.
gld_returns <- gld_prices_clean %>%
  mutate(
    log_return = 100 * (log(adjusted) - log(lag(adjusted))),
    simple_return = 100 * ((adjusted / lag(adjusted)) - 1)
  ) %>%
  drop_na(log_return, simple_return)

# Flag possible return outliers using Tukey's IQR rule.
Q1 <- quantile(gld_returns$log_return, 0.25, na.rm = TRUE)

```

```

Q3 <- quantile(gld_returns$log_return, 0.75, na.rm = TRUE)
IQR_val <- Q3 - Q1
lower_fence <- Q1 - 1.5 * IQR_val
upper_fence <- Q3 + 1.5 * IQR_val

gld_returns <- gld_returns %>%
  mutate(return_outlier_iqr = log_return < lower_fence | log_return > upper_fence)

# Optional sensitivity dataset: winsorized returns.
winsorize <- function(x, lower_prob = 0.01, upper_prob = 0.99) {
  lo <- quantile(x, lower_prob, na.rm = TRUE)
  hi <- quantile(x, upper_prob, na.rm = TRUE)
  pmin(pmax(x, lo), hi)
}

gld_returns_winsorized <- gld_returns %>%
  mutate(log_return_winsorized = winsorize(log_return))

# Save starting datasets
write.csv(gld_prices_clean, file.path(out_dir, "gld_daily_prices_clean.csv"),
  row.names = FALSE)
write.csv(gld_returns, file.path(out_dir, "gld_daily_returns.csv"), row.names =
  FALSE)
write.csv(gld_returns_winsorized, file.path(out_dir,
  "gld_daily_returns_winsorized_sensitivity.csv"), row.names = FALSE)

# Data summary table
summary_table <- tibble(
  ticker = symbol,
  start_date = min(gld_prices_clean$date),
  end_date = max(gld_prices_clean$date),
  price_rows = nrow(gld_prices_clean),
  return_rows = nrow(gld_returns),
  raw_duplicate_dates = raw_duplicate_dates,
  total_raw_missing_values = sum(raw_missing_counts),
  iqr_return_outliers_flagged = sum(gld_returns$return_outlier_iqr),
  mean_adjusted_close = mean(gld_prices_clean$adjusted),
  min_adjusted_close = min(gld_prices_clean$adjusted),
  max_adjusted_close = max(gld_prices_clean$adjusted),
  mean_daily_log_return = mean(gld_returns$log_return),
  sd_daily_log_return = sd(gld_returns$log_return)
)

write.csv(summary_table, file.path(out_dir, "data_summary_table.csv"), row.names =
  FALSE)

```

```

print(summary_table)
cat("\nMissing values by row column:\n")
print(raw_missing_counts)

# =====
# 2. Data visualizations
# =====

p_price <- ggplot(gld_prices_clean, aes(x = date, y = adjusted)) +
  geom_line() +
  labs(
    title = "GLD Adjusted Closing Price Over Time",
    x = "Date", y = "Adjusted Close Price (USD)"
  ) +
  theme_minimal()

ggsave(file.path(out_dir, "plots", "01_gld_adjusted_price.png"), p_price, width =
9, height = 5)

p_returns <- ggplot(gld_returns, aes(x = date, y = log_return)) +
  geom_line() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(
    title = "GLD Daily Log Returns Over Time",
    x = "Date", y = "Daily Log Return (%)"
  ) +
  theme_minimal()

ggsave(file.path(out_dir, "plots", "02_gld_log_returns_time_series.png"),
p_returns, width = 9, height = 5)

p_hist <- ggplot(gld_returns, aes(x = log_return)) +
  geom_histogram(bins = 60) +
  labs(
    title = "Histogram of GLD Daily Log Returns",
    x = "Daily Log Return (%)", y = "Count"
  ) +
  theme_minimal()

ggsave(file.path(out_dir, "plots", "03_gld_log_returns_histogram.png"), p_hist,
width = 8, height = 5)

p_box <- ggplot(gld_returns, aes(y = log_return)) +
  geom_boxplot() +
  labs(
    title = "Boxplot of GLD Daily Log Returns",

```

```

    y = "Daily Log Return (%)"
  ) +
  theme_minimal()

ggsave(file.path(out_dir, "plots", "04_gld_log_returns_boxplot.png"), p_box, width
= 6, height = 5)

# Base R diagnostic plots saved as PNG
png(file.path(out_dir, "plots", "05_acf_pacf_returns.png"), width = 1000, height =
700)
par(mfrow = c(1, 2))
acf(gld_returns$log_return, main = "ACF: GLD Log Returns")
pacf(gld_returns$log_return, main = "PACF: GLD Log Returns")
par(mfrow = c(1, 1))
dev.off()

png(file.path(out_dir, "plots", "06_acf_pacf_squared_returns.png"), width = 1000,
height = 700)
par(mfrow = c(1, 2))
acf(gld_returns$log_return^2, main = "ACF: Squared GLD Returns")
pacf(gld_returns$log_return^2, main = "PACF: Squared GLD Returns")
par(mfrow = c(1, 1))
dev.off()

png(file.path(out_dir, "plots", "07_qq_plot_returns_diagonal_plot.png"), width =
800, height = 700)
qqnorm(gld_returns$log_return, main = "Normal Q-Q Plot of GLD Daily Log Returns")
qqline(gld_returns$log_return, lwd = 2)
dev.off()

# =====
# 3. Time-series train/test split
# =====
# Use chronological split, not random split, because time-series forecasting must
train on the past and test on the future.
# A 70/30 split matches the structure we used in the other essays.
n <- nrow(gld_returns)
train_n <- floor(0.70 * n)
train_data <- gld_returns[1:train_n, ]
test_data <- gld_returns[(train_n + 1):n, ]

train_ret <- train_data$log_return
test_ret <- test_data$log_return

split_table <- tibble(

```

```

dataset = c("Training set", "Test set"),
rows = c(nrow(train_data), nrow(test_data)),
start_date = c(min(train_data$date), min(test_data$date)),
end_date = c(max(train_data$date), max(test_data$date))
)
write.csv(split_table, file.path(out_dir, "train_test_split_table.csv"), row.names
= FALSE)
print(split_table)

# =====
# 4. Assumption / preliminary tests
# =====
# ADF test: checks stationarity of returns. For GARCH, returns should usually be
stationary.
adf_result <- adf.test(train_ret)

# Ljung-Box test on returns: checks autocorrelation in mean.
lb_returns <- Box.test(train_ret, lag = 20, type = "Ljung-Box")

# Ljung-Box test on squared returns and ARCH test: checks volatility clustering /
ARCH effects.
lb_squared_returns <- Box.test(train_ret^2, lag = 20, type = "Ljung-Box")
arch_result <- ArchTest(train_ret, lags = 12)

prelim_tests <- tibble(
  test = c("ADF stationarity test", "Ljung-Box returns", "Ljung-Box squared
returns", "ARCH LM test"),
  statistic = c(
    unname(adf_result$statistic),
    unname(lb_returns$statistic),
    unname(lb_squared_returns$statistic),
    unname(arch_result$statistic)
  ),
  p_value = c(
    adf_result$p.value,
    lb_returns$p.value,
    lb_squared_returns$p.value,
    arch_result$p.value
  )
)
write.csv(prelim_tests, file.path(out_dir, "preliminary_tests.csv"), row.names =
FALSE)
print(prelim_tests)

# =====

```

```

# 5. ARMA mean model selection
# =====
# We use BIC because it favors a simpler model.
arma_fit_auto <- auto.arima(
  train_ret,
  max.p = 5,
  max.q = 5,
  d = 0,
  seasonal = FALSE,
  stationary = TRUE,
  ic = "bic",
  stepwise = FALSE,
  approximation = FALSE
)

print(arma_fit_auto)
arma_order <- arma_fit_auto$arma
arma_p <- arma_order[1]
arma_q <- arma_order[2]
cat("\nSelected ARMA order by BIC: ARMA(", arma_p, ", ", arma_q, ")\n", sep = "")

```